POWER BI PROJECT REPORT

Smart Energy Management & Sustainability Analytics

Executive Portfolio Overview And Store Efficiency Ranking & Audit Matrix

60	44.79M	₹45.4 Cr	11.8%
Stores analysed	Units (kWh) used	Electricity spend	Covered by solar

Tools used: Python, Pandas, SQL (MySQL) ,Power BI for dashboards.

Dashboards: Energy Management & Cost Optimization Dashboard | Store Efficiency Ranking & Audit Matrix

1. What this project is about

Imagine a company with **60 stores** across India: big hypermarkets, supermarkets, department stores and small convenience stores. Every store uses electricity for lights, air conditioning, freezers and billing counters, and every month the company receives an electricity bill.

This project takes all that data (daily meter readings, monthly bills, solar power produced, store sizes and cities) and turns it into **two easy-to-read Power BI dashboards** that show where the money and energy are going.

2. The problem

The company uses about **44.79 million units (kWh) of electricity** and pays about **₹45.4 crore**. Nobody could easily answer these questions:

- Which stores are wasting electricity?
- Which cities and store types cost the most?
- In which months do we use the most power?
- Are small and large stores being compared fairly?
- Are bills being paid on time, and how much does solar help?

Without answers, audits were done at random and savings were missed. The dashboards give clear answers in one place.

3. Problems we solved

Sl.no	Problem	Solved on
1	See the big numbers (usage, cost, efficiency, warnings) at a glance	Page 1
2	Find which stores need a check-up first	Pages 1 & 2
3	Compare small and big stores fairly	Page 2
4	Find which cities use and spend the most	Pages 1 & 2
5	Find which type of store uses the most electricity	Page 1
6	Find the busiest and quietest months	Page 1
7	Check how many bills are paid or still pending	Page 2
8	See how much solar helps and what the price per unit is	Page 2
9	Give auditors one table with a clear "High-Risk" flag	Page 2

4. The two dashboards

Name of Dashboard	Energy Management & Cost Optimization Dashboard	Store Efficiency Ranking & Audit Matrix
Who uses it	Managers and leadership	Auditors and maintenance teams
What it tells you	What is happening and when	Which stores have a problem and why

5. Two terms to know

EUI (Energy Usage Intensity)
Electricity used **per square foot** of store. Think of it like a car's mileage: it lets you compare a tiny shop and a giant hypermarket fairly. **Lower is better.**

Z-score
Shows how **unusual** a day's electricity use is. **0 is normal; above 2 is unusually high** and counts as a warning. The higher the score, the bigger the problem.

6. Page 1: Energy Management & Cost Optimization

The "big picture" page for managers.

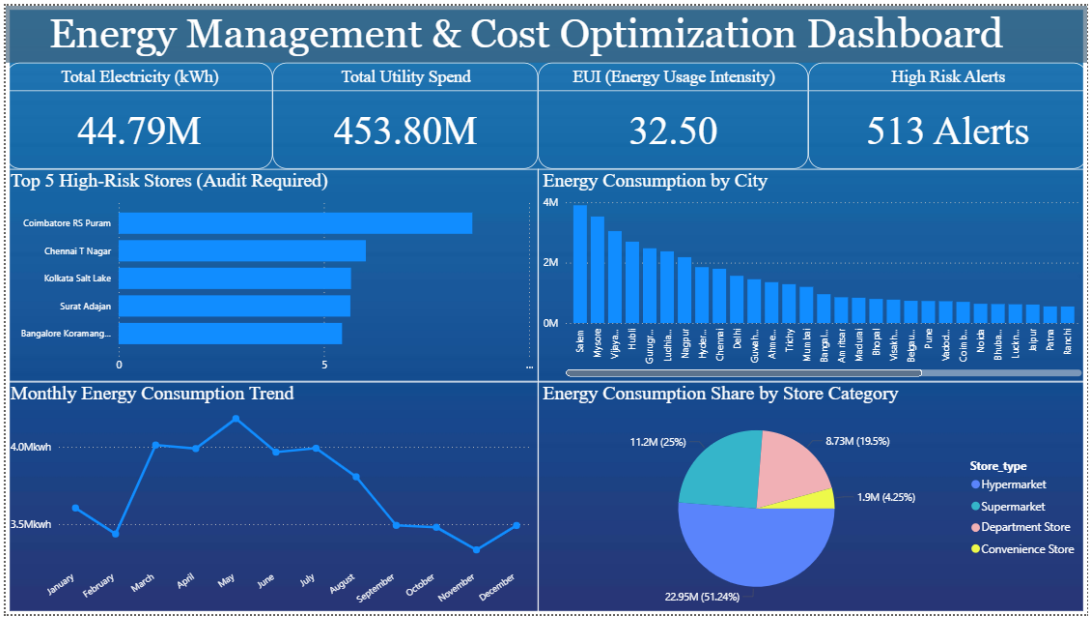


Figure 1: Energy Management & Cost Optimization Dashboard

Charts used and what they solve

Power BI visual	What it shows	Problem it solves
Four number cards (KPI cards)	44.79M units used, ₹45.38 crore spent, efficiency 32.5, 513 warnings	Headline numbers for leadership in seconds
Bar chart: Top 5 High-Risk Stores	The five stores with the most unusual usage	Tells the audit team where to go first
Column chart: Energy by City	Electricity used per city, highest to lowest	Shows which locations drive usage
Line chart: Monthly Trend	Electricity use from January to December	Reveals busy and quiet months
Pie chart: Share by Store Type	How much each store type uses	Shows which format uses the most

What Page 1 tells us

1. Hypermarkets use half of everything. They take 51% of all electricity. Supermarkets use 25%, department stores 19.5% and convenience stores only 4%. The three bigger formats account for about 96%.

Store type	Units used	Share
Hypermarket	22.95M	51.24%
Supermarket	11.2M	25%
Department Store	8.73M	19.5%
Convenience Store	1.9M	4.25%

2. Summer is the expensive season. Usage climbs from February, **peaks in May**, stays high until July, and is lowest in November. The peak month uses about a quarter more than the quietest one, pointing to air conditioning and refrigeration working harder in the heat.

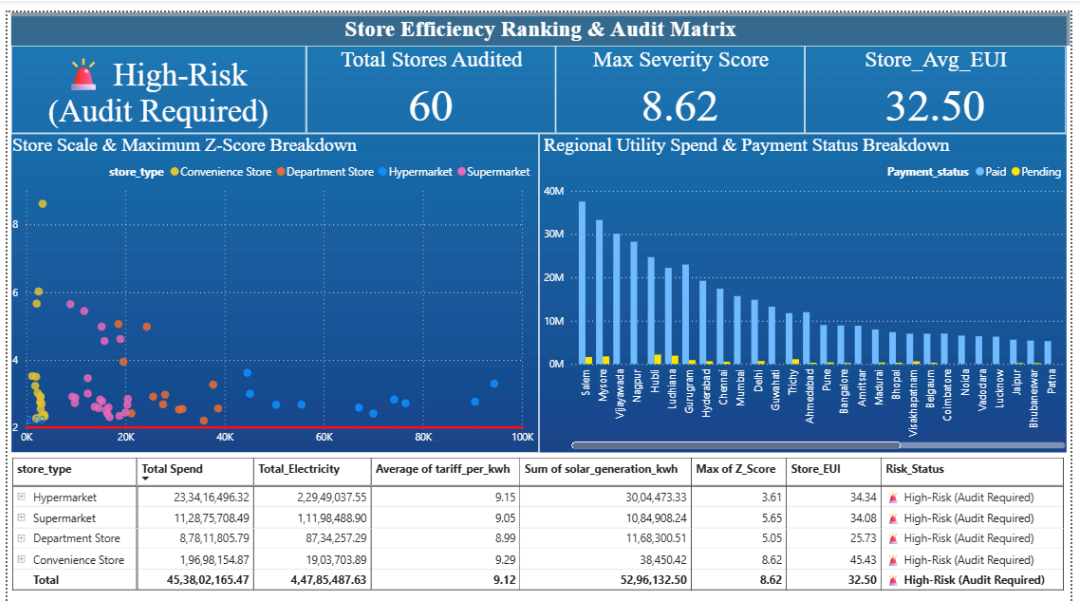
3. A few cities matter most. Salem leads (about 3.9M units), followed by Mysore (about 3.5M) and Vijayawada (about 3.05M). After that, usage drops steadily across many smaller cities, so savings efforts should begin with the top few.

4. Warnings are frequent. 513 unusually-high-usage alerts show the problem happens throughout the year, not on just a few bad days.

5. The riskiest stores are not in the biggest cities. The top five are Coimbatore RS Puram, Chennai T Nagar, Kolkata Salt Lake, Surat Adajan and Bangalore Koramangala. Risk comes from a store using far more than similar stores, not from being large. Coimbatore RS Puram stands out most.

7. Page 2: Store Efficiency Ranking & Audit Matrix

The "detective" page for auditors: which stores have a problem, and how bad is it?



store_type	Total Spend	Total_Electricity	Average of tariff_per_kwh	Sum of solar_generation_kwh	Max of Z_Score	Store_EUI	Risk_Status
Hypermarket	23,34,16,496.32	2,29,49,037.55	9.15	30,04,473.33	3.61	34.34	High-Risk (Audit Required)
Supermarket	11,28,75,708.49	1,11,98,488.90	9.05	10,84,908.24	5.65	34.08	High-Risk (Audit Required)
Department Store	8,78,11,805.79	87,34,257.29	8.99	11,68,300.51	5.05	25.73	High-Risk (Audit Required)
Convenience Store	1,96,98,154.87	19,03,703.89	9.29	38,450.42	8.62	45.43	High-Risk (Audit Required)
Total	**45,38,02,165.47**	**4,47,85,487.63**	**9.12**	**52,96,132.50**	**8.62**	**32.50**	**High-Risk (Audit Required)**

Figure 2: Store Efficiency Ranking & Audit Matrix

Charts used and what they solve

Power BI visual	What it shows	Problem it solves
Four number cards (KPI cards)	High-Risk flag, 60 stores checked, worst score 8.62, average efficiency 32.5	Audit coverage and how bad the worst case is
Dot chart (scatter plot) with a red line at 2	Each dot is a store: size on one side, unusual-usage score on the other	Any dot above the red line is a problem, whatever its size
Column chart: Spend by City	Money spent per city, split into Paid and Pending bills	Where the money goes and what is unpaid
Table (matrix)	Per store type: cost, usage, price per unit, solar, worst score, efficiency, risk flag	One view for auditors to decide

What Page 2 tells us

1. Small convenience stores are the least efficient.

Store type	Efficiency (lower is better)	Worst score	Share of electricity
Convenience Store	45.4	8.6	4%
Hypermarket	34.3	3.6	51%
Supermarket	34.1	5.7	25%
Department Store	25.7	5.1	19.5%
Whole chain	32.5	8.6	100%

Convenience stores use only 4% of the electricity, yet use about 40% more per square foot than the chain average. On the dot chart they are the small stores with the highest scores. This points to things like old equipment, freezers running poorly or lights left on.

2. Hypermarkets are big but steady. They have the highest bill (₹23.3 crore, about half of the total) but the lowest warning score (3.6). Their usage is consistent, so the saving comes from using less overall, not from fixing a one-off fault.

3. Department stores are the most efficient. They are about 21% better than the chain average and can serve as the benchmark.

4. Price per unit is not the problem. The tariff is about ₹9.12 per unit in every store type (between ₹9.0 and ₹9.3). Since the price is almost the same everywhere, **negotiating a better rate will not save much. The saving has to come from using less.**

5. Solar covers about 12% of usage. The company produces about 53 lakh units from solar, covering **11.8%** of its electricity.

Store type	Solar covers
Department Store	~13%
Hypermarket	~13%
Supermarket	~10%
Convenience Store	~2%

Convenience stores are inefficient and have almost no solar, so they are the best place to add it.

6. Late payment is not the problem. In every city the unpaid (pending) part is small compared with the paid part. Salem, Mysore, Vijayawada, Nagpur and Hubli spend the most, matching the usage on Page 1. The largest pending amounts are in Hubli, Ludhiana, Mysore and Salem, so follow up there first.

8. Conclusion

- **Page 1 gives the big picture:** the company uses 44.79 million units and spends ₹45.4 crore, usage peaks in summer, and hypermarkets use half of everything.
- **Page 2 shows where the problems are:** small convenience stores are the least efficient, and Coimbatore RS Puram is the most unusual store.

> **The main finding**
> The cost problem comes from **how much electricity is used**, not from the price per unit or late payments. The solution is to **use less and generate more of our own.**

What we recommend

1. **Audit the top 5 high-risk stores first**, starting with Coimbatore RS Puram, then check convenience stores in general.
2. **Cut waste in hypermarkets and supermarkets.** They use about 76% of the electricity, so even small savings add up.
3. **Prepare for summer.** Service air conditioning and refrigeration before March.
4. **Add solar panels**, especially at convenience stores.
5. **Follow up on pending bills** in Hubli, Ludhiana, Mysore and Salem.